PROJECT OVERVIEW

Modeling Sensory Dissonance of Musical Intervals

Background & motivation

A musical interval consists of two notes played together. Intervals can be described as consonant or dissonant. A chord that sounds "pleasant" or "agreeing" is consonant, whereas a chord that sounds "unpleasant" or "tense" is dissonant.

The frequency ratio between the two notes in an interval will be denoted by ρ.

The study of consonance is said to have begun with Pythagoras (Bowling and Purves). His theory, along with some later ones, generally agree that intervals with simple frequency ratios (e.g. $2/1, 3/2, \ldots$) are consonant, whereas those with complicated ratios (e.g. $45/32$) are dissonant (Bowling and Purves).

Intervals formed by the Western 12-tone music scale are shown in table 1. This table associates intervals to three categories, from most consonant to least consonant, according to conventional consensus: perfect consonance (PC) > imperfect consonance (IC) > dissonance (D).

Table 1: Intervals formed by the 12-tone Western music scale (Müller 241 and "Intervals in the Major Scale").

Interval Name	**Abbreviation**	**Frequency ratio ρ (higher freq. / lower freq.)**	**Consonance / dissonance**
Perfect unison	P1	$1/1$	PC
Minor second	m2	$16/15$	D
Major second	M2	$9/8$	D
Minor third	m3	$6/5$	IC
Major third	M3	$5/4$	IC
Perfect fourth	P4	$4/3$	PC
Aug. fourth / dim. fifth	A4 or d5	$45/32$	D
Perfect fifth	P5	$3/2$	PC
Minor sixth	m6	$8/5$	IC
Major sixth	M6	$5/3$	IC
Minor seventh	m7	$9/5$	D
Major seventh	M7	$15/8$	D
Perfect octave	P8	$2/1$	PC
$\vdots$	$\vdots$	$\vdots$	$\vdots$

However, some intervals are formed from notes that are not in the 12-tone scale. Conventional theories do not tell us what happens when frequency ratios fall in between those in table 1. Such instances are common in traditional/folk music, as well as contemporary microtonal music.

Thus, we seek to develop a method that allows us to do the following:

- Given a recording of an interval, assign a value that represents the consonance/dissonance of that interval.
- Given a recording of one note, define a function that outputs the dissonance $D(\rho)$ for the interval formed by that note and another note with the same timbre, at frequency ratio ρ.

Pure tones

When simple vibration occurs alone at one particular frequency, the resulting pure tone can be represented by a sinusoid, defined as follows.

A pure tone is a sound consisting of periodic pressure oscillations at a particular frequency. We can also define a pure tone $s_{\text{pure}}(t)$ as a signal

$$s_{\text{pure}}(t) = a\sin(2\pi ft + \phi)$$

where $s_{\text{pure}}(t)$ represents air pressure relative to atmospheric pressure, at a given point in space at time t. In this definition, a is the amplitude, f is the frequency and $1/f$ is the period, ϕ is the phase, relative to the instant $t = 0$.

The properties in the above definition are shown in figure 1.

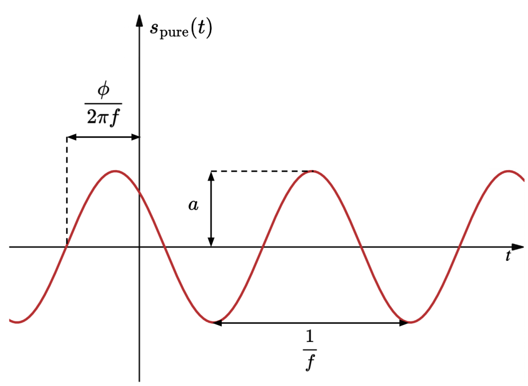

Figure 1: Graph of a pure tone $s_{\text{pure}}(t) = a\sin(2\pi ft + \phi)$

Beating, roughness & sensory dissonance

When multiple pure tones sound simultaneously, they superpose, which means the resultant signal is the sum the pure tones.

For example, we can express the signal for the sum of pure tones s_1 and s_2 of different frequency as follows:

$$(s_1 + s_2)(t) = a_1 \sin(2\pi f_1 t + \phi_1) + a_2 \sin(2\pi f_2 t + \phi_2).$$

This situation with two pure tones is represented in figure 2.

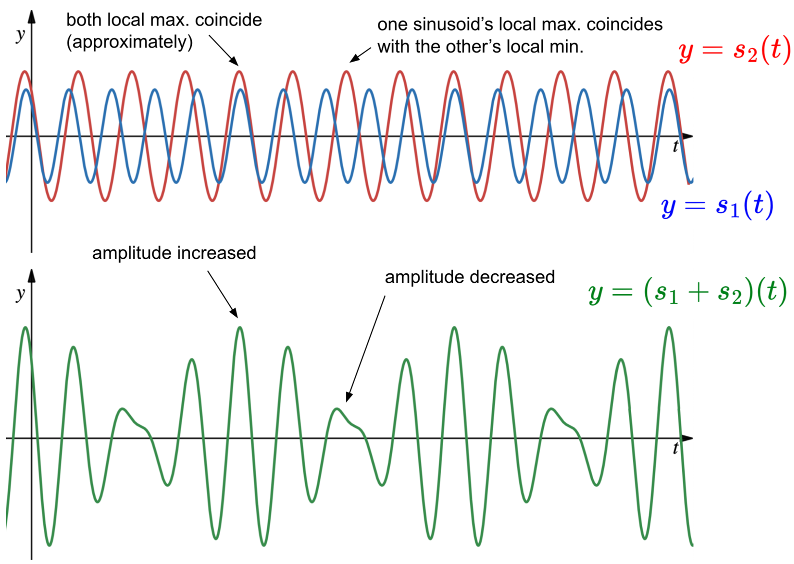

Figure 2: Two pure tones (top) and their sum—the resultant signal (bottom) The scale of both graphs is the same.

We observe that at some instants, the pure tones are in phase, meaning the local maxima and local minima of one wave align with those of the other respectively. This increases the resultant amplitude.

At other instants, the pure tones are out of phase, meaning the local maxima of one wave align with local minima of the other. This decrease the resultant amplitude.

The oscillation between constructive and destructive interference causes periodic increases and decreases in the volume of the resulting sound. Each oscillation in volume is called a beat, and the phenomenon is called beating (Sethares 40–41). Roughness refers to the effect caused by rapid beating in a sound (Sethares 46).

Beating and roughness are perceived as unpleasant and cause dissonance (Helmholtz 226). Of course, there are other factors affecting how a sound is perceived, including cultural factors. This is why we distinguish sensory dissonance from dissonance in general.

Sensory dissonance refers to the dissonance caused by beating and roughness (Sethares 62–65).

We can quantify dissonance according to the definition above: the more frequently beats occur, and the larger the variation in resultant amplitude, the more dissonance. Hence, it seems natural to try to quantify dissonance as a positive value, rather than consonance.

Beat amplitude is the difference between the maximum and minimum amplitudes of a signal in a beat.

Beat frequency and beat amplitude

The two following variables allow us to define a dissonance function: beat frequency and beat amplitude.

Beat frequency F is the number of beats that occur per unit time.

We can deduce the beat frequency as follows. This method is from Sethares (327). Consider the sum of two pure tones s_1 and s_2, each of amplitude 1:

$$(s_1 + s_2)(t) = \sin(2\pi f_1 t + \phi_1) + \sin(2\pi f_2 t + \phi_2),$$

and assume $f_1 \geq f_2$ since the equation is symmetric with respect to s_1 and s_2.

Applying the sum-product identity

$$\sin(X) + \sin(Y) = 2\cos\left(\frac{X-Y}{2}\right)\sin\left(\frac{X+Y}{2}\right),$$

we can write

$$(s_1 + s_2)(t) = 2\cos\left[2\pi\left(\frac{f_1 - f_2}{2}\right)t + \frac{\phi_1 - \phi_2}{2}\right]\sin\left[2\pi\left(\frac{f_1 + f_2}{2}\right)t + \frac{\phi_1 + \phi_2}{2}\right].$$

Letting $\Delta f = f_1 - f_2$, $\Delta\phi = \phi_1 - \phi_2$, $\Sigma f = f_1 + f_2$ and $\Sigma\phi = \phi_1 + \phi_2$, we have

$$(s_1 + s_2)(t) = \underbrace{2\cos\left[2\pi\left(\frac{\Delta f}{2}\right)t + \frac{\Delta\phi}{2}\right]}_{s_A(t)}\sin\left[2\pi\left(\frac{\Sigma f}{2}\right)t + \frac{\Sigma\phi}{2}\right].$$

In other words, $s_A(t) = 2\cos\left[2\pi\left(\Delta f/2\right)t + \Delta\phi/2\right]$ modulates the amplitude of a higher-frequency sinusoid, which precisely describes the phenomenon of beating (see Figure 3).

The function $s_A(t)$ has frequency $\Delta f/2$. Over each period of s_A, there are two beats, so the beat frequency is double the frequency of s_A, that is, $F = \Delta f = f_1 - f_2$.

The maximum amplitude of a signal $(s_1 + s_2)(t)$ is reached at instants of maximum constructive interference. Amplitudes of s_1 and s_2 add to each other, so maximum amplitude is $a_1 + a_2$.

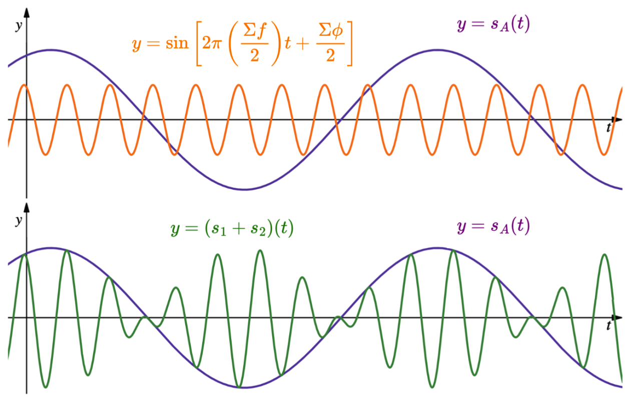

Figure 3: Visualizing beating as the modulation of a sinusoid by another sinusoid.

Minimum amplitude is reached at instants of maximum destructive interference. Amplitudes of s_1 and s_2 subtract from each other, making the resultant amplitude $|a_1 - a_2|$.

So for a tone of the form $(s_1 + s_2)(t) = a_1 \sin(2\pi f_1 t + \phi_1) + a_2 \sin(2\pi f_2 t + \phi_2)$, the beat amplitude is

$$A = a_1 + a_2 - |a_1 - a_2|.$$

Dissonance function for intervals of pure tones

Eventually, when frequency difference F exceeds a certain threshold, we begin to perceive two completely separate pitches with less dissonance, rather than beating/roughness on a single pitch (Costa 3). This means there is not just a simple increasing relationship between beat frequency and dissonance.

A more realistic representation of sensory dissonance accounts for this (Plomp and Levelt). The shape of this curve is shown in figure 4. This particular curve represents the sensory dissonance created by a pure tone of varying frequency (x-axis) played simultaneously with another pure tone of frequency fixed at $400\,\mathrm{Hz}$ (Sethares 47).

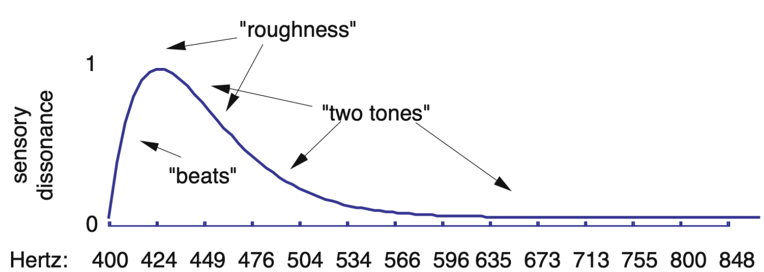

Figure 4: Realistic dissonance curve (Sethares 47).

To represent the dissonance d of two pure tones, we will use the form $d = \mathrm{e}^{-pF} - \mathrm{e}^{-qF}$ which is one provided by Sethares, and which corresponds to this type of dissonance curve (345). We will also use Sethares's parameters $p = 0.035,\ \ q = 0.0575$, which were optimized based on Plomp and Levelt data (Sethares 345).

We should also consider beat amplitude in our dissonance function. We can do this by multiplying $\left(\mathrm{e}^{-pF} - \mathrm{e}^{-qF}\right)$ by beat amplitude A. We can now define our function.

For an interval of pure tones of the form $(s_1 + s_2)(t) = a_1 \sin(2\pi f_1 t + \phi_1) + a_2 \sin(2\pi f_2 t + \phi_2)$, we define the dissonance function d as follows:

$$d(f_1, f_2, a_1, a_2) = A(\mathrm{e}^{-pF} - \mathrm{e}^{-qF}) = \left(a_1 + a_2 - |a_1 - a_2|\right)\left(\mathrm{e}^{-p|f_1-f_2|} - \mathrm{e}^{-q|f_1-f_2|}\right)$$

with parameters $p = 0.035$ and $q = 0.0575$.

'Complex' tones

When playing on a real instrument, different portions of the instrument vibrate at different frequencies. This produces multiple pure tones and results in a complex[1] tone, even when playing a single note. Each pure tone produced by an instrument, and contained within the resultant complex tone, is called a partial of that complex tone.

Complex tones, like pure tones, are periodic, but contain more than one partial. In other words, a complex tone is a sum of multiple pure tones. The signal representing a complex tone is written as

$$s_{\text{complex}}(t) = \sum_{j=1}^{L} s_j(t) \quad \text{where} \quad s_j(t) = a_j \sin(2\pi f_j t + \phi_j).$$

In this expression, s_j is the jth partial of s_{complex}, and L is the total number of partials.

Since we now need to think about all the different frequencies contained in a tone, what we're really interested in is the frequency spectrum of the tone .

From the physics of waves, the *fundamental* frequency refers to the frequency of the lowest partial (f), and higher *harmonic* partial frequencies are integer multiples of the fundamental. In general, amplitude of partials decreases at higher frequencies. Frequency spectra vary between instruments, as shown in figure 5.

Dissonance function for complex tones

When two complex tones are played simultaneously, their frequency spectra overlap to form a combined frequency spectrum, as shown in figure 6.

[1] Note that the word 'complex' here has nothing to do with complex numbers. Instead, 'complex tones' are just tones with more than one frequency component.

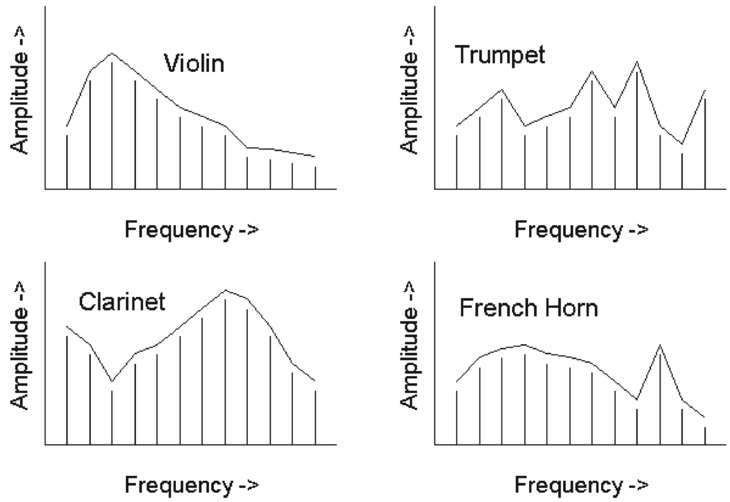

Figure 5: Sketches of frequency spectra of different musical instruments (Devi).

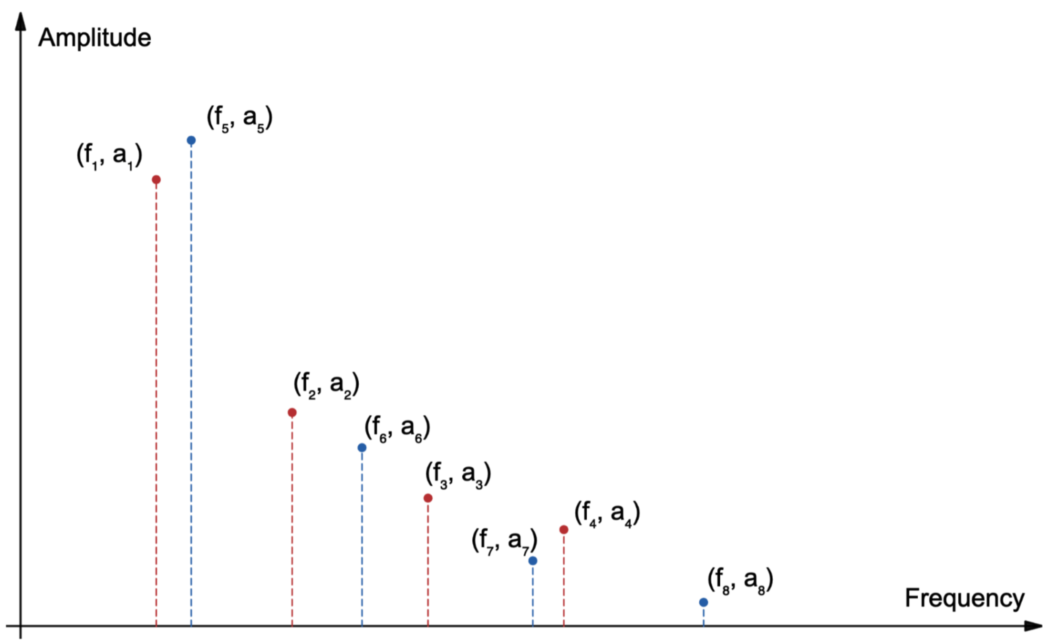

Figure 6: Example combined frequency spectrum. Partials of lower tone (f_1, a_1) to (f_4, a_4) shown in red and partials of higher tone (f_5, a_5) to (f_8, a_8) shown in blue.

This gives us a new combined spectrum. Each pair of partials produces dissonance, so to obtain the overall dissonance value, we can sum the dissonance values for each combination of pure tones contained in the new spectrum (Sethares 346). Thus, we can finally write the following formula, similar to Sethares (346).

We define the dissonance value D for intervals of complex tones as

$$D = \sum_{j=1}^{L} \sum_{r=j+1}^{L} d(f_j, f_r, a_j, a_r) = \sum_{j=1}^{L} \sum_{r=j+1}^{L} \left(a_j + a_r - |a_j - a_r|\right)\left(\mathrm{e}^{-p|f_j - f_r|} - \mathrm{e}^{-q|f_j - f_r|}\right) \qquad (1)$$

where $f_1, f_2, \cdots, f_L$ are the frequencies of partials in the combined frequency spectrum, $a_1, a_2, \cdots, a_L$ are the amplitudes of partials in the combined spectrum, L is the total number of partials in the combined spectrum, and $p = 0.035$, $q = 0.0575$.

Computing frequency spectra

Now we know how to quantify the dissonance of an interval based on the amplitudes and frequencies of all the partials in each of the tones. Thus, we just need a way to obtain the frequency spectrum of tones based on a sound recording. We can do this using the Discrete Fourier Transform (DFT).

A sound recording is essentially a vector $\boldsymbol{x} = \begin{bmatrix} x_0 & x_1 & \cdots & x_{N-1} \end{bmatrix}^T$ containing N samples, where x_k is the value of the signal recorded at the time t_k. The instant t_k can be expressed as $t_k = k/f_s$ where f_s is the sampling rate.

The DFT of such a signal $\boldsymbol{x}$ is given as

$$\mathbf{X} = \begin{bmatrix} X_0 & X_1 & \cdots & X_{N-1} \end{bmatrix}^T \quad \text{where} \quad X_k = \sum_{n=0}^{N-1} x_n \mathrm{e}^{-i2\pi \xi_k t_n}.$$

Then $|X_k|$ is related to the amplitude of the partial of frequency $\xi_k = k f_s / N$.

A frequency spectrum can be obtained by plotting $|X_k|$ values as a function of ξ_k.

Dissonance should be independent of the loudness of tones in the recording, so we will normalize the amplitude values before inputting them into our dissonance formula (equation 1). We will do this by dividing them by the largest amplitude in their respective note/tone, $|X|_{\max}$. So for a given partial, our normalized amplitude is $a = |X|/|X|_{\max}$.

Illustration of the method

To summarize on our method for determining the consonance of intervals:

1. Record the interval being played.
2. Compute the normalized frequency spectrum of the interval and identify the frequencies and amplitudes of partials involved.
3. Compute the dissonance value of the interval according to equation 1, by plugging in amplitude and frequency values from step 2.

Dissonance value of one interval

We recorded the following signal of a *major second* interval on a violin over $1.6\mathrm{s}$. This consists of two tones, which we will call Ψ_1 and Ψ_2. MATLAB was used to compute the frequency spectra (shown in figure 7) using the FFT function. We were then able to extract he data in tables 2 and 3, and obtain a dissonance value of $D = 0.451$. This result does not mean much, since we have nothing to compare it to. To generate a whole range of dissonance values that can be compared with each other, we can use dissonance curves.

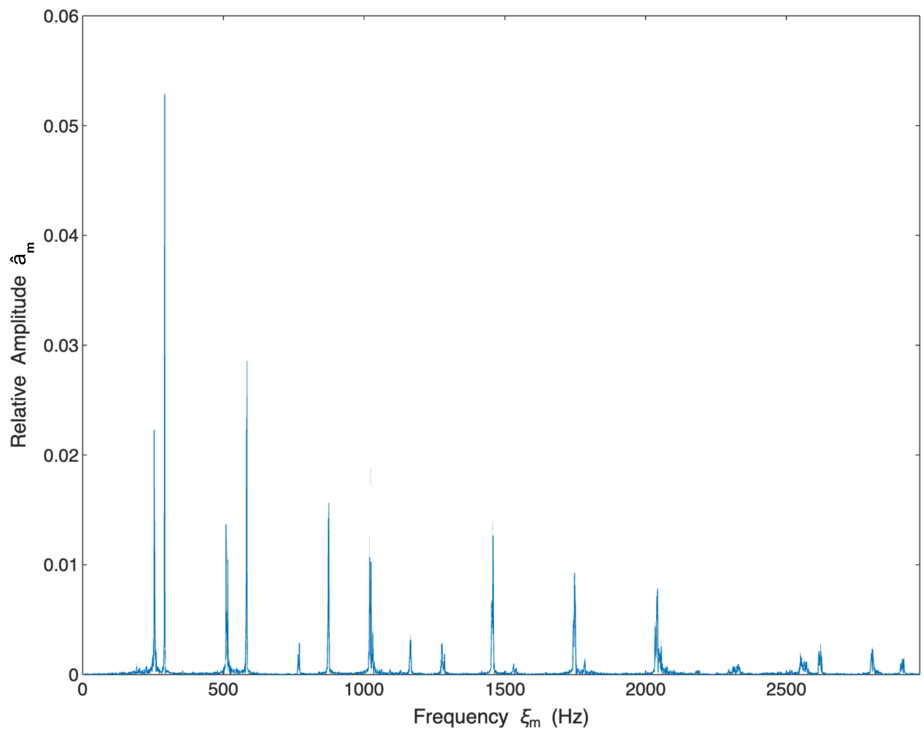

Figure 7: Frequency Spectrum. Amplitudes are also on a relative scale, so there are no units on the $\hat{a}_m$-axis.

Table 2: Coordinates of peaks corresponding to first 8 partials of each tone.

ξ_m	$\lvert X_m \rvert$		ξ_m	$\lvert X_m \rvert$
254.372	0.0222587		1276.86	0.00289803
291.246	0.0528551		1456.23	0.0141559
509.368	0.0136538		1531.86	0.000949981
583.117	0.0285483		1748.10	0.0092127
769.365	0.0025358		1784.35	0.00136827
874.364	0.0156332		2034.35	0.00434093
1019.36	0.0106837		2042.47	0.00782107
1164.98	0.00340102		2331.22	0.000916981

Table 3: Normalized amplitudes and frequencies of partials of each tone in the recording.

Partials of lower tone Ψ_1					Partials of higher tone Ψ_2			
j	f_j	$\|X_j\|$	a_j		j	f_j	$\|X_j$—	a_j
1	254.372	0.0222587	1.000		9	291.246	0.0528551	1.000
2	509.368	0.0136538	0.613		10	583.117	0.0285483	0.540
3	769.365	0.0025358	0.114		11	874.364	0.0156332	0.296
4	1019.36	0.0106837	0.480		12	1164.98	0.00340102	0.064
5	1276.86	0.00289803	0.130		13	1456.23	0.0141559	0.268
6	1531.86	0.000949981	0.043		14	1748.10	0.0092127	0.174
7	1784.35	0.00136827	0.061		15	2042.47	0.00782107	0.148
8	2034.35	0.00434093	0.195		16	2331.22	0.000916981	0.017

Dissonance Curves

We will now consider the interval formed by Ψ_2 and a new tone, Ψ_ρ, which has the same relative amplitudes as Ψ_2, except with all frequencies multiplied by an arbitrary ratio ρ (see figure 8). From now on, let $f_1, f_2, \cdots$ and $a_1, a_2, \cdots$ be the partial frequencies and amplitudes of Ψ_2.

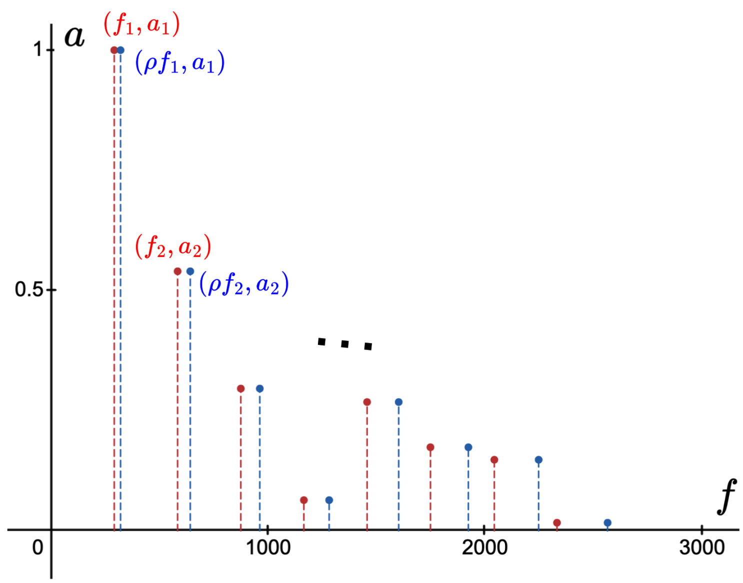

Figure 8: Frequency spectrum of Ψ_2 (red) and that of Ψ_ρ (in blue). In this figure, $\rho = 1.1$.

We can now determine how the dissonance of the interval of Ψ_2 and Ψ_ρ changes as the ratio ρ changes. Treating D as a function of ρ, we obtain the curve in figure 9.

This dissonance curve agrees fairly well with conventional notions outlined in table 1 in the introduction. Notably, all perfect consonances are at local minima of $D(\rho)$, whereas traditional dissonances are generally higher up on the graph compared to neighboring consonances.

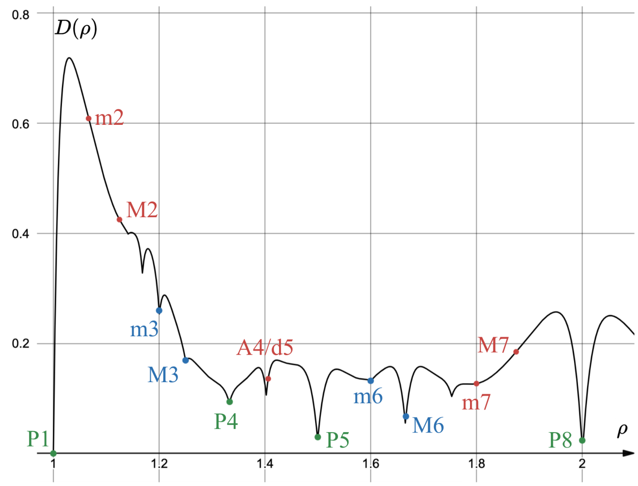

Figure 9: Dissonance curve for Ψ_2 and Ψ_ρ. Labelled with abbreviated interval names from table 1 in the introduction. Perfect consonances in green, imperfect consonances in blue, and dissonances in red. Figure created using Desmos.

Works Cited

"A Musician's Intuition on the Fourier Transform (Feat. The Inner Product) — 3b1b SoME1." *Youtube*, 7 Aug. 2021, youtube.com/watch?v=ToMyB5Hk06w. Accessed 26 July 2024.

Bowling, Daniel, and Dale Purves. "A Biological Rationale for Musical Consonance." *Proceedings of the National Academy of Sciences*, vol. 112, no. 36, Sept. 2015, https://doi.org/10.1073/pnas.1505768112. Accessed 12 Mar. 2024.

Brunton, Steven. "The Discrete Fourier Transform (DFT)." *Youtube*, 28 Mar. 2020, youtube.com/watch?v=nl9TZanwbBk. Accessed 31 July 2024.

Costa, Luciano Da Fontoura. *Modeling Consonance and Its Relationships with Temperament and Harmony (CDT-10)*. HAL, 2019, hal.science/hal-02151895/document. Accessed 7 Mar. 2024.

Devi, J. Sirisha. "A Study : Analysis of Music Features for Musical Instrument Recognition and Music Similarity Search." *International Journal of Computer Science and Informatics*, vol. 2, no. 3, Jan. 2013, https://doi.org/10.47893/ijcsi.2013.1088. Accessed 12 May 2024.

"Fourier Transforms." *MathWorks*, mathworks.com/help/matlab/math/fourier-transforms.html. Accessed 4 Feb. 2024.

Helmholtz, Hermann von. *On the Sensations of Tone as a Physiological Basis for the Theory of Music*. 2nd ed., Longmans, Green, 1885, books.google.ca/books?redir_esc=y&id=GwE6AAAAIAAJ&q=. Accessed 26 Apr. 2024.

"Intervals in the Major Scale". Carnegie Mellon University, andrew.cmu.edu/user/johnito/music_theory/fundamentals/Intervals.pdf. Accessed 15 Aug. 2024.

Kameoka, Akio, and Mamoru Kuriyagawa. "Consonance Theory Part I: Consonance of Dyads." *The Journal of the Acoustical Society of America*, vol. 45, no. 6, 1969, https://doi.org/10.1121/1.1911623. Accessed 15 Aug. 2024.

Meinard Müller. *Fundamentals of Music Processing: Audio, Analysis, Algorithms, Applications*. 1st ed., Springer, 2015.

Plomp, R., and W. J. M. Levelt. "Tonal Consonance and Critical Bandwidth." *The Journal of the Acoustical Society of America*, vol. 38, no. 4, 1965, https://doi.org/10.1121/1.1909741.

Sethares, William. *Tuning, Timbre, Spectrum, Scale*. 2nd ed., Springer, 2005, r-5.org/files/books/rx-music/tuning/William_A_Sethares-Tuning_Timbre_Spectrum_Scale-EN.pdf. Accessed 15 Aug. 2024.

Smith, Julius. *Mathematics of the Discrete Fourier Transform (DFT) : With Audio Applications*. 2nd ed., W3K, 2007.

"Why Sound Wave Is Called a Longitudinal Wave?" *Toppr*, Haygot Technologies, toppr.com/ask/question/give-reason-for-the-following-why-sound-wave-is-called-a-longitudinal-wave/. Accessed 15 Aug. 2024.